9.3 Conditional Probability

Name _____ Period ____ Date _____

Algebra I 9.3 · Student sheet

I can…

- find a conditional probability from a sample space or a two-way table.
- use the formula for conditional probability.
- use conditional probability to solve real-life problems.

Vocabulary · fill in the missing word

Conditional probability — the probability that event B happens given that event A has _____, written P(B | A)

P(B | A) — read "the _____ of B given A"

Formula — P(B | A) = _____

Part 1 · Shade the grid

The grid shows every sum when two dice are rolled (first die down the side, second die across the top).

1st \ 2nd	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

a. Shade every sum of 9. P(sum is 9) = _____

b. Cover every row except the row where the first die is 4. P(sum is 9 | first die is 4) = _____

c. Why did the probability change? _____

Part 2 · Notes: fill in the blanks

P(B | A) means the probability of _____, given that _____ has happened.

"Given" shrinks the _____ to the outcomes in A.

From a table: P(B | A) = (number in A and B) ÷ _____

Formula: P(B | A) = _____ ÷ _____

Part 3 · Examples (work them with the class)

Example 1

A bag holds 3 red marbles and 2 blue marbles. You choose one marble, keep it, then choose another. Given that the first marble is red, what is the probability that the second is blue?

Example 2

A survey of 100 students asked about exercise and sleep.

a. Find P(8+ hours of sleep | exercises).

b. Find P(exercises | 8+ hours of sleep).

	8+ hours sleep	Less	Total
Exercises	36	24	60
Doesn't	14	26	40
Total	**50**	**50**	**100**

Example 3

At a school, 60% of the students take Spanish, and 24% take Spanish and art. What is the probability that a student who takes Spanish also takes art?

Example 4

In a survey of 100 people, 30 own only a dog, 15 own only a cat, 10 own both, and 45 own neither. What is the probability that a dog owner also owns a cat?

Part 4 · Partner practice

Partner 1

Use the exercise and sleep table. Find P(doesn’t exercise | less than 8 hours of sleep).

	8+ hours sleep	Less	Total
Exercises	36	24	60
Doesn’t	14	26	40
Total	**50**	**50**	**100**

Partner 2

85% of students pass the first test, and 70% pass both tests. What is the probability that a student passes the second test, given that the student passed the first?